

WARRNAMBOOL, Australia

WMO: 948370

Lat: 38.2867S

Lon: 142.4522E

Elev: 71

StdP: 100.47

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.2	3.4	1.1	4.1	4.8	2.1	4.4	5.6	13.6	10.5	12.8	10.5	3.1	10	0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.0	32.7	19.3	29.5	18.6	26.6	18.0	20.8	28.0	19.7	26.6	18.7	25.0	6.9	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	13.6	22.5	17.5	12.6	21.3	16.6	11.9	20.5	60.5	27.9	56.5	26.7	53.3	25.1	33.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.4	11.6	10.3	-0.1	39.2	0.9	2.7	-0.8	41.1	-1.3	42.6	-1.8	44.2	-2.5	46.1		
			-0.3	23.2	0.9	1.6	-1.0	24.3	-1.5	25.2	-2.0	26.1	-2.6	27.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.2	17.2	17.8	16.4	14.1	11.8	9.4	9.8	10.8	12.3	14.0	15.6		
		DBStd	4.05	3.74	3.62	3.32	2.79	2.20	1.78	1.63	1.93	2.16	2.92	3.36		
		HDD10.0	108	0	0	0	1	6	24	32	27	12	6	1		
		HDD18.3	1989	67	49	82	132	202	254	278	264	224	191	140		
		CDD10.0	1285	224	219	198	123	63	19	12	21	38	76	120		
		CDD18.3	126	33	34	22	5	1	0	0	0	2	9	19		
		CDH23.3	1462	393	342	231	61	3	0	0	1	44	136	251		
		CDH26.7	622	191	154	93	16	0	0	0	0	10	43	115		
(14)	Wind	WSAvg	5.2	5.4	5.2	4.9	4.6	5.0	4.8	5.5	5.3	5.3	5.2	5.3		
(15)	Precipitation	PrecAvg	774	34	30	48	61	74	82	93	96	81	71	60		
		PrecMax	1016	112	92	153	140	202	168	162	203	161	138	113		
		PrecMin	487	4	0	9	8	15	18	41	18	33	15	2		
		PrecStd	129	21	24	27	36	37	33	31	36	30	33	27		
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.2	35.7	33.7	29.0	22.1	17.1	15.9	19.0	22.4	28.0	31.0		
			MCWB	20.4	19.9	18.6	16.5	14.7	13.6	12.2	14.0	15.8	18.3	19.1		
		2%	DB	31.8	31.7	29.1	24.7	19.0	15.1	14.3	16.5	19.3	23.8	27.3		
			MCWB	19.4	19.9	18.2	15.6	14.0	12.5	11.6	12.5	14.3	17.0	18.2		
		5%	DB	27.6	27.6	25.5	21.6	17.2	14.0	13.3	14.8	17.1	20.8	23.9		
			MCWB	18.5	18.8	17.6	15.1	13.6	11.8	10.9	11.6	13.1	15.6	17.4		
		10%	DB	23.8	24.1	22.2	19.0	15.7	13.1	12.5	13.4	15.3	18.0	20.5		
			MCWB	17.7	17.9	16.5	14.5	12.9	11.2	10.4	10.9	12.2	14.1	16.1		
		0.4%	WB	22.4	22.7	20.8	17.9	16.1	14.6	13.2	14.6	16.3	19.5	20.9		
			MCDB	30.3	29.0	27.2	25.1	18.7	15.7	14.7	18.3	20.6	25.2	27.7		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	20.6	20.6	19.1	16.7	14.9	13.1	12.2	13.0	14.7	17.4	19.1		
			MCDB	28.2	27.6	26.2	21.6	17.6	14.5	13.6	15.6	18.4	22.9	24.9		
		5%	WB	19.2	19.4	18.1	15.8	14.1	12.2	11.4	12.0	13.6	15.8	17.6		
			MCDB	26.1	26.2	23.9	20.0	16.4	13.5	12.8	14.2	16.5	20.5	22.9		
		10%	WB	18.0	18.2	17.1	14.9	13.2	11.4	10.7	11.2	12.5	14.3	16.1		
			MCDB	23.4	23.4	21.4	18.4	15.3	12.8	12.1	13.0	14.9	17.4	20.5		
		5% DB	MDBR	10.7	10.0	9.7	9.3	7.4	6.8	6.4	7.1	8.2	9.5	9.8		
			MCDBR	18.5	16.7	16.3	14.6	10.3	8.1	7.5	9.5	11.7	15.6	16.6		
		5% WB	MCWBR	7.4	6.6	6.9	7.0	6.4	5.9	5.4	6.3	7.4	9.1	8.6		
			MCWBR	15.2	13.7	13.3	11.9	8.7	7.0	6.7	8.4	10.5	14.5	15.0		
(40)	Clear-Sky Solar Irradiance	taub	0.346	0.340	0.329	0.317	0.309	0.302	0.299	0.312	0.336	0.341	0.342	0.337		
		taud	2.494	2.529	2.560	2.578	2.577	2.581	2.586	2.541	2.454	2.448	2.475	2.512		
		Ebn at Noon	986	968	939	888	833	809	836	880	914	956	984	1000		
		Edh at Noon	113	105	94	80	69	64	68	82	102	112	115	112		
(44)	All-Sky Solar Radiation	RadAvg	7.13	6.13	4.83	3.36	2.23	1.77	1.95	2.80	4.02	5.38	6.28	7.06		
		RadStd	0.40	0.33	0.24	0.22	0.11	0.13	0.12	0.15	0.25	0.38	0.37	0.43		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.39	N/A	N/A	+1.06	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	+0.45	N/A	N/A	+0.87	+0.39	N/A	N/A	-117	+150	+40

Nomenclature: See separate page